Lab Sheet-04

Object Oriented Programming Using Java

LastDate:16-10-2026

MCA III Semester

1. **Student Record**
   Create a Student class with name and roll number. The main class creates the object and calls `display()`. No display logic is written in main.
2. **Medicine**
   Create Medicine with a parameterized constructor. Use `this` to initialize instance variables.
3. **Pharmacy**
   Create Pharmacy with a static pharmacy name and location. Demonstrate that the static value is shared.
4. **Pharmacy Inventory**
   Create MedicineStock objects in an array and display them using a loop. The array and loop belong to Inventory, not main.
5. **Patient Eligibility**
   Use an if-else condition inside a Patient method to classify a patient.
6. **Encapsulated Medicine Product**
   Make medicine data private. The class itself displays its data.
7. **Bank Account**
   Control changes to a private bank balance through a deposit method with validation.
8. **Patient Bill**
   Use private fees with setters and a total calculation method. No calculation is placed in main.
9. **Employee → Pharmacist**
   Create a parent Employee and child Pharmacist. Use single inheritance.
10. **Hierarchical Hospital Staff**
    Create one Employee parent and two child classes: Doctor and Pharmacist.
11. **Multilevel Hospital Staff**
    Create Person → Employee → Manager. Manager receives behavior through the chain.
12. **Multiple Inheritance: Concept Only**
    Understand why Java cannot extend two classes. Do not use interfaces yet.
13. **Hybrid Inheritance: Concept Only**
    Understand a hybrid structure without implementing multiple class inheritance.
14. **Shape Hierarchy**
    Create Shape as parent and Rectangle/Circle as children. Each child calculates its own area.
15. **Final Hospital Staff System**
    Integrate the allowed concepts in one small application. Main only creates the system object and calls its method.